

■ AI Lesson for Kids ■

Overfitting:

When AI Memorises Instead of Learns

■ Subject: AI & Digital Literacy

■ Level: Beginner (Ages 8–14)

■ Time: ~45 minutes

■ Learning Objectives

1. Understand what Artificial Intelligence (AI) is and how it learns.
2. Explain the difference between **learning** and **memorising**.
3. Define **overfitting** in simple, everyday language.
4. Identify real-life examples of overfitting.
5. Describe why overfitting is a problem for AI systems.
6. Recognise the signs that an AI might be overfitting.

■ Section 1 — What is Artificial Intelligence?

Artificial Intelligence — or **AI** — is technology that lets computers think and learn a bit like humans do. Instead of being told exactly what to do every time, an AI looks at lots of examples and figures out the **pattern** by itself.

Think of AI as a very fast learner that reads millions of books, pictures, or conversations in just seconds. After all that reading it can answer questions, recognise faces, translate languages, and even beat humans at chess!

Where do we find AI every day?

■ Games	Game characters that react to your moves
■ Phones	Face unlock and voice assistants like Siri
■ Cars	Self-driving features and parking helpers
■ Music	Spotify / YouTube recommending songs you love
■ Medicine	Scanning X-rays to spot health problems

■ **Key Idea:** AI learns *patterns* from examples, just like you learn to ride a bike by practising — not by reading every rule about balance!

■ Section 2 — Learning vs Memorising

Before we understand overfitting, we need to know the difference between **truly learning** something and just **memorising** it. These might sound the same, but they are very different!

■ Real Learning	■ Just Memorising
Understands the <i>meaning</i> behind facts	Remembers the exact words or answer
Can solve problems it has never seen before	Can only repeat what it already saw
Adapts when things change slightly	Gets confused if anything is different
Gets smarter and more flexible over time	Stays stuck with the same limited knowledge
Like understanding <i>why</i> $2+2=4$	Like memorising ' $2+2=4$ ' with no idea why

A classroom story: Imagine two students — Aarav and Priya — studying for a maths test. Aarav memorises every question and answer from last year's paper. Priya understands *how* to solve each type of problem.

On test day the teacher uses brand-new questions. Aarav is stuck — none of the questions match what he memorised! Priya solves them all because she understands the method. **Priya learned; Aarav only memorised.**

■ Section 3 — What is Overfitting?

Overfitting happens when an AI learns its training data *too perfectly*. Instead of picking up general patterns it can use on new data, it essentially memorises every tiny detail — including random mistakes and noise that aren't really part of the true pattern.

■ **Definition:** Overfitting is when an AI model memorises the training examples so closely that it fails to work correctly on new, unseen examples.

How training works (in simple steps):

Step 1	Engineers collect lots of examples (called training data).
Step 2	The AI studies those examples and looks for patterns.
Step 3	The AI is tested on new data it has never seen before.
Step 4	If it does well on new data → great! The AI has learned .
Step 5	If it does well only on old data → problem! The AI has overfitted .

■ Section 4 — Real-Life Examples of Overfitting

■ The Cat Example

An AI is trained to recognise cats using 1,000 photos — all orange tabbies sitting on a blue sofa. The AI memorises 'orange fur + blue sofa = cat'. When shown a black cat on a red carpet it says 'NOT a cat!' — because it memorised details instead of learning what makes a cat a cat.

■ The Exam Example

A student memorises every answer from five practice papers word for word. In the real exam, the wording is slightly different. The student fails even though the ideas are the same. The student overfitted to the practice papers.

■ The Weather Example

An AI weather app is trained only on data from one city. It memorises the exact patterns of that city. When used in another city it gives completely wrong forecasts — it never learned general weather rules, only that one city's quirks.

■ The Music Example

A music-recommendation AI is trained on one person's playlist for a week. It memorises those exact songs. When a different user with different tastes tries it, the AI recommends the wrong music entirely.

■ Section 5 — Why is Overfitting a Problem?

Overfitting makes AI **less useful in the real world**. Here is why that matters:

■ Medical AI

If a disease-detection AI overfits, it might miss real illnesses in patients whose scans look slightly different from the training scans.

■ Self-Driving Cars	An overfitted driving AI might recognise roads it trained on but get confused by new roads, weather, or signs.
■ Security Systems	A face-recognition system that overfits might fail to identify real users or mistakenly let strangers in.
■ Business Decisions	Companies use AI to predict sales. An overfitted model might predict perfectly for last year but get next year completely wrong.

■ Section 6 — How to Spot Overfitting

Scientists and engineers watch out for these **warning signs**:

■ Does well on training data	■ But does badly on new data
■ Answers are too specific	■ Can't handle even small changes
■ 100% accurate in practice runs	■ Makes lots of mistakes in real life
■ Memorises exact details	■ Misses the big picture / general rule

■ **How do we fix overfitting?** Engineers use tricks like giving the AI **more varied data**, adding rules that stop it memorising tiny details (called **regularisation**), and always testing on data the AI has **never seen before**.

■ Quick Summary

AI	A smart computer system that learns patterns from data.
Learning	Understanding the <i>why</i> so you can solve new problems.
Memorising	Storing exact answers without understanding the pattern.
Training Data	The examples the AI studies to learn from.
Overfitting	When AI memorises training data too well and fails on new data.
The Fix	More varied data, testing on new examples, and regularisation.

■ Multiple-Choice Questions (20 Questions)

■ **Instructions:** Read each question carefully and circle the letter of the best answer. There is only **one correct answer** per question. Take your time and think before you answer!

Q1. What does AI stand for?

- A) Automatic Input
- B) Artificial Intelligence
- C) Amazing Internet
- D) Automatic Information

Q2. Which of the following is the best description of AI?

- A) A robot that can only walk
- B) A computer program that learns patterns to solve problems
- C) A calculator that does maths
- D) A video game with colourful graphics

Q3. Which of these is NOT an example of AI in everyday life?

- A) A voice assistant on your phone
- B) A self-driving car
- C) A pencil used to write
- D) A music app that recommends songs you like

Q4. What is the main difference between learning and memorising?

- A) Learning is faster than memorising
- B) Memorising means you can solve any new problem
- C) Learning means understanding ideas so you can handle new problems
- D) There is no difference between them

Q5. A student memorises the exact answers to last year's exam paper but does not understand the topic. What will happen on a new exam?

- A) They will get full marks
- B) They will struggle because the new questions are different
- C) They will learn everything quickly
- D) They will teach the class

Q6. What is 'training data' in AI?

- A) Data that the AI deletes
- B) Data that is kept secret from the AI
- C) Examples the AI studies to learn patterns
- D) Data that makes the AI tired

Q7. What is overfitting?

- A) When an AI learns too slowly
- B) When an AI memorises training data too perfectly and fails on new data
- C) When an AI is given too many tasks
- D) When an AI refuses to answer questions

Q8. An AI trained only on orange cats cannot recognise a black cat. This is an example of:

- A) Good learning
- B) Overfitting
- C) Underfitting
- D) Perfect training

Q9. Which sign suggests that an AI might be overfitting?

- A) It does well on both training data and new data
- B) It does perfectly on training data but badly on new data
- C) It refuses to process any data
- D) It learns new data faster than old data

Q10. Why is overfitting a problem for a medical AI?

- A) It makes the AI run faster
- B) It helps the AI remember every patient
- C) It might miss real illnesses in patients whose scans look slightly different
- D) It makes diagnosis cheaper

Q11. Which of the following best describes what a well-trained AI should do?

- A) Memorise every example it was shown
- B) Understand patterns well enough to handle new, unseen examples
- C) Stop learning after one day
- D) Repeat only the exact answers it has seen

Q12. Priya understands how to solve maths problems; Aarav only memorised answers. In AI terms, Aarav is like an AI that has:

- A) Learned correctly
- B) Overfitted to the training data
- C) Been given more data
- D) Used regularisation

Q13. What kind of data would help fix overfitting in a cat-recognition AI?

- A) More photos of only orange cats on blue sofas
- B) No photos at all
- C) A wide variety of cat photos — different colours, sizes, and backgrounds
- D) Photos of dogs instead

Q14. What does it mean when we say an AI 'generalises well'?

- A) The AI can only answer general trivia questions
- B) The AI performs well on new data it has never seen before
- C) The AI takes a very long time to answer
- D) The AI knows every fact in the world

Q15. Which of these is a technique used to reduce overfitting?

- A) Giving the AI less data
- B) Never testing the AI on new data
- C) Using regularisation to stop the AI memorising tiny details
- D) Making the AI answer faster

Q16. An overfitted weather AI was trained only on one city's data. What will happen when it is used in a different city?

- A) It will give correct forecasts
- B) It will give completely wrong forecasts because it memorised one city's quirks
- C) It will ask for more data
- D) It will learn the new city instantly

Q17. What is 'noise' in training data?

- A) Sounds that come from the computer
- B) Random or irrelevant details that are not part of the true pattern
- C) Extra useful information for the AI
- D) Music files given to the AI

Q18. If an AI does well only on examples it has already seen, we say it has poor _____.

- A) Memory
- B) Speed
- C) Generalisation
- D) Colour recognition

Q19. Which statement about overfitting is TRUE?

- A) Overfitting makes AI more useful in the real world
- B) Overfitting only happens in video games
- C) An overfitted AI struggles when problems are slightly different from its training
- D) Overfitting is always good for self-driving cars

Q20. What is the best way to check if an AI has overfitted?

- A) Ask the AI how it feels
- B) Count how many examples it was trained on
- C) Test the AI on new data it has never seen before and see if it still performs well
- D) Make the AI faster

■ Answer Key

Q1: B	Q2: B	Q3: C	Q4: C
Q5: B	Q6: C	Q7: B	Q8: B
Q9: B	Q10: C	Q11: B	Q12: B
Q13: C	Q14: B	Q15: C	Q16: B
Q17: B	Q18: C	Q19: C	Q20: C

Score

Name: _____

Total score: _____ / 20

■ Keep curious and keep learning about AI! ■